

Keeping Them Cool: Thermal-Hydraulics of Superconducting Cables and Magnets

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Spring semester 2026

LECTURE 6 (SC)



Let's move a step backward...

Multiphysics Modeling of Superconducting Cables for Fusion

Integrating Electromagnetism, Cryogenic Materials, and
Thermal-Hydraulics

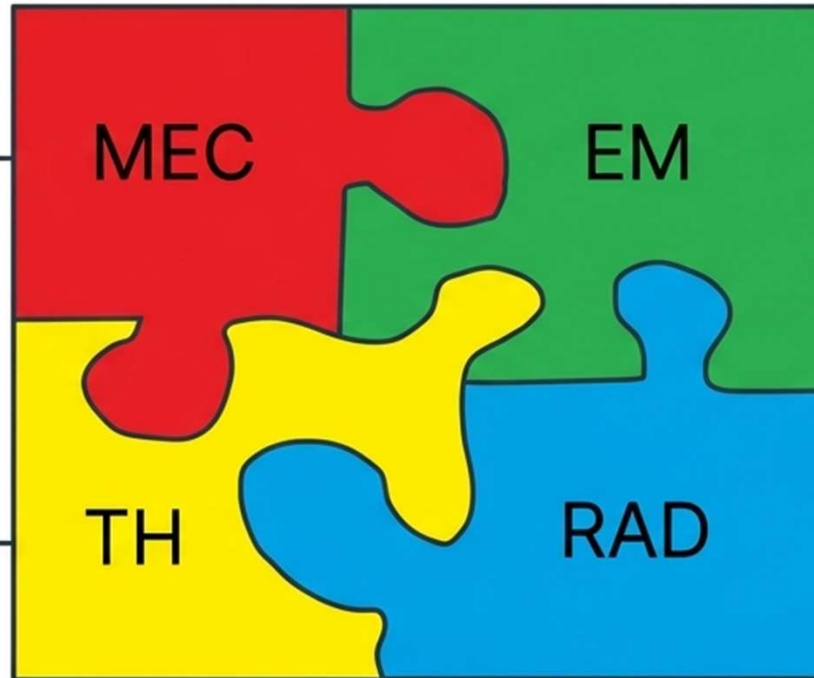
The Multiphysics Imperative

Mechanics & Materials:

Surviving massive Lorentz forces and cryogenic material embrittlement.

Thermal-Hydraulics:

Maintaining the superconducting state via active, supercritical refrigeration.



Electromagnetism:

Managing high critical current densities (J_c) and massive magnetic fields (H_c).

Radiation: Managing external nuclear heat loads from the fusion plasma.

Superconducting cables for fusion cannot be designed in silos. A failure in one domain (e.g., inadequate TH cooling) immediately triggers catastrophic failure in another (loss of EM superconducting state).

OUTLINE



PART 1: SC magnet cooling strategies

- Alternative solutions
- Properties at cryogenic temperature

PART 2: Active cooling for SC conductors

- Cable-In-Conduit Conductor (CICC) concept
- Layouts of LTS and HTS CICC for fusion (and power transmission)
- (beyond the Cable-In-Conduit concept)

PART 3: Numerical modelling of CICC thermal-hydraulics

- Timescale and spatial scale separation
- Governing equations
- A look to the winding pack and cooling circuit



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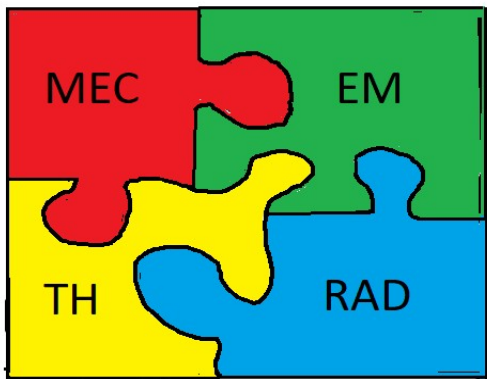
Part 3

NUMERICAL MODELLING OF CICC THERMAL-HYDRAULICS

MAHTEP Group, Energy Department "Galileo Ferraris", Politecnico di Torino

NELAHTP - Spring semester 2026

Refrigeration of LTS cable-in-conduit conductors with SHe at 4.5 K



Intersecting Physics

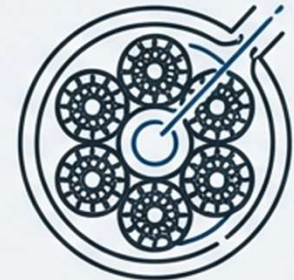
Multi-physics nature of the problem (Thermal-Hydraulics + Electromagnetic + Mechanical).
(+ Radiation)



Extreme Scale Spans

Multi-scale temporal domain: 10^{-3} s to 10^2 s.

Multi-scale spatial domain: 10^{-6} m to 10^2 m.

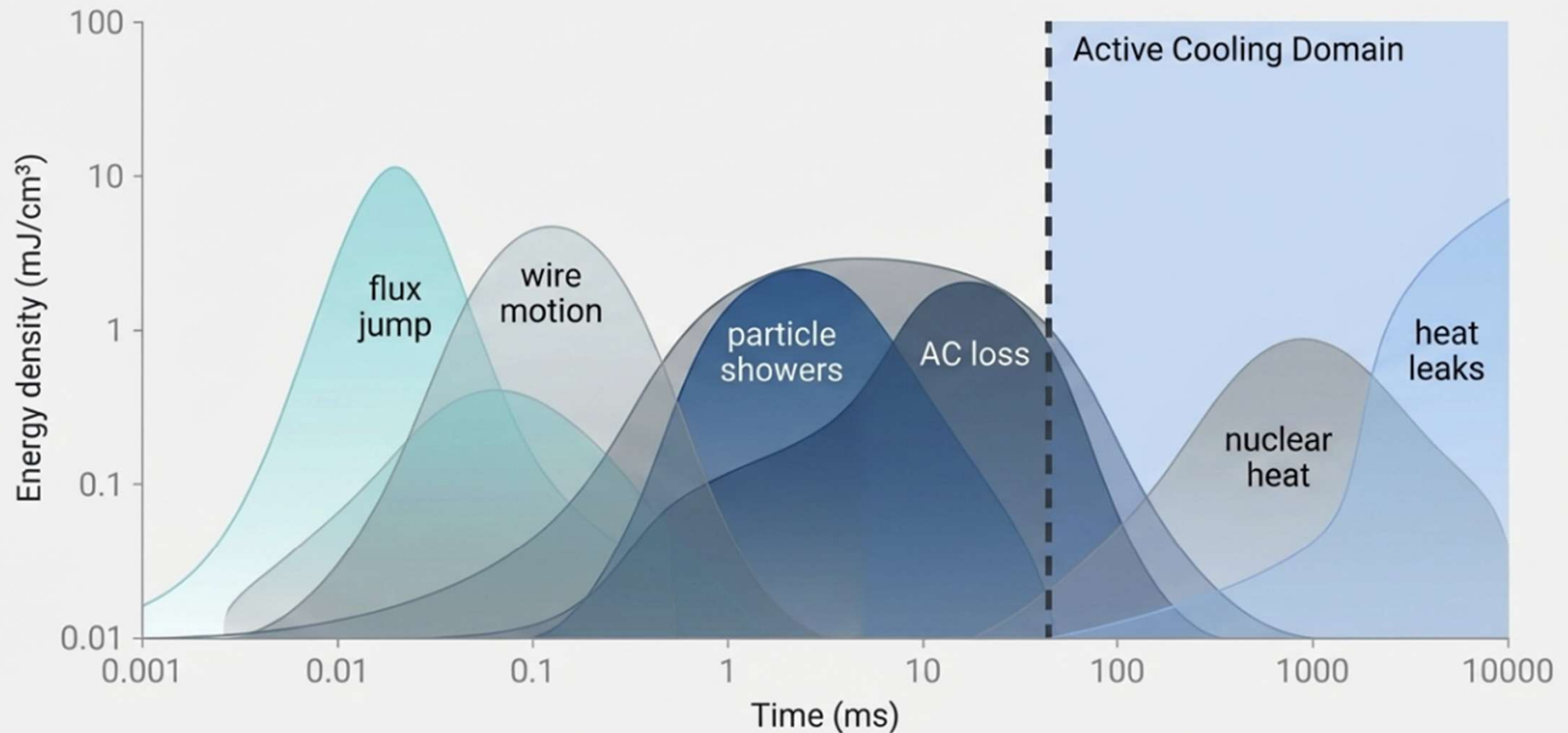


Structural Complexity

Complex cable bundle structure and interaction between constituents.

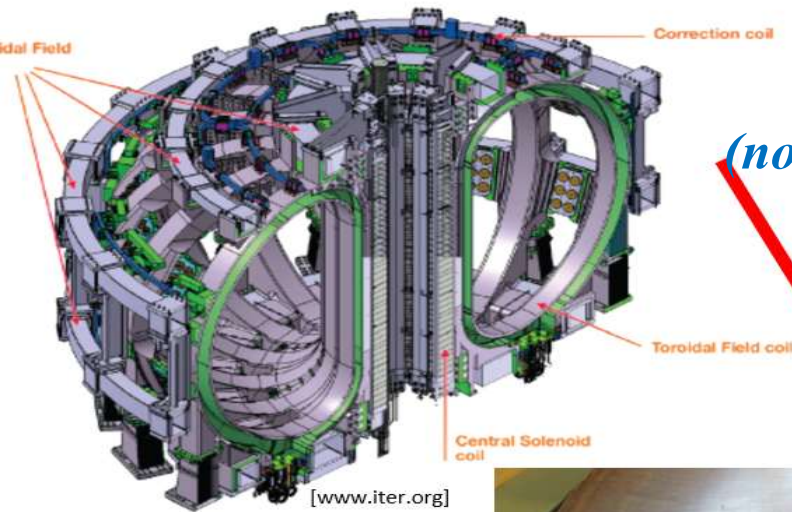
Small temperature margin!

Time scales: Active cooling acts on timescales $> 10\text{-}100\text{ ms}$.



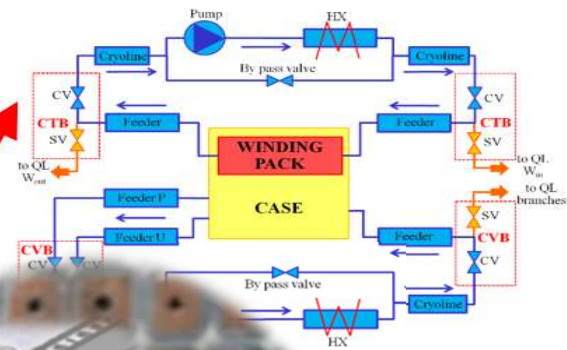
Spatial Scales: A Multi-Scale Topology

MACRO ($10^1 - 10^2$ m)

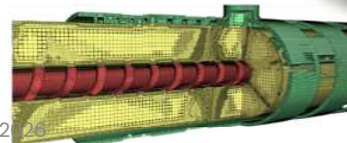
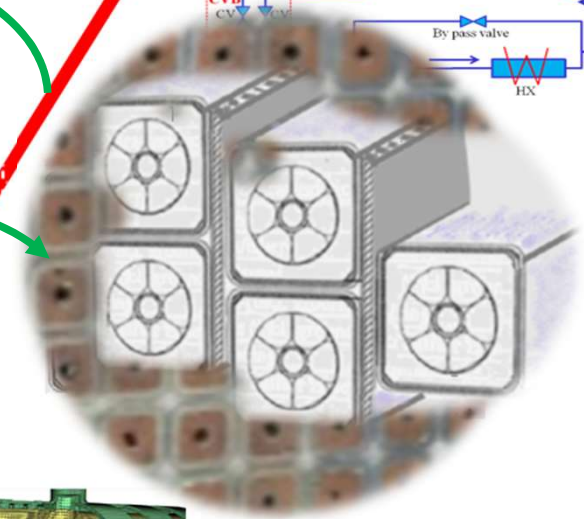


Functional analysis
(normal operation, safety analyses)

MACRO ($10^1 - 10^2$ m)



MESO ($10^{-2} - 1$ m)



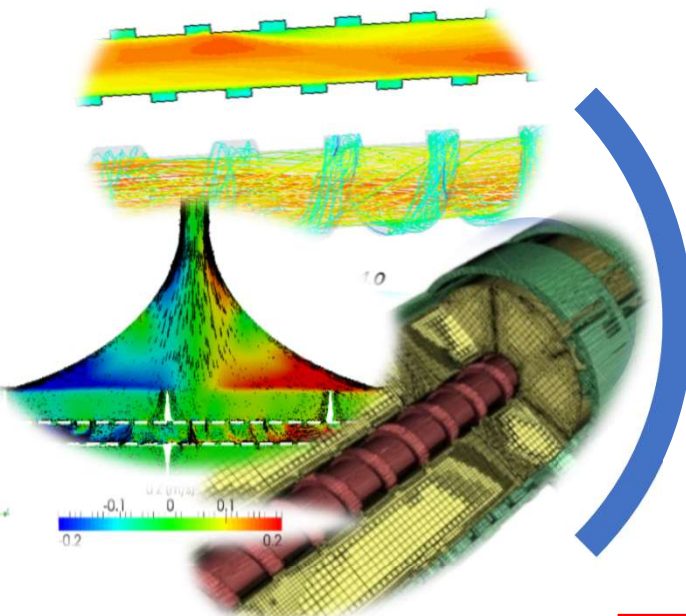
ENELAHTP - 1st year semester 2026

MACRO ($10^1 - 10^2$ m)

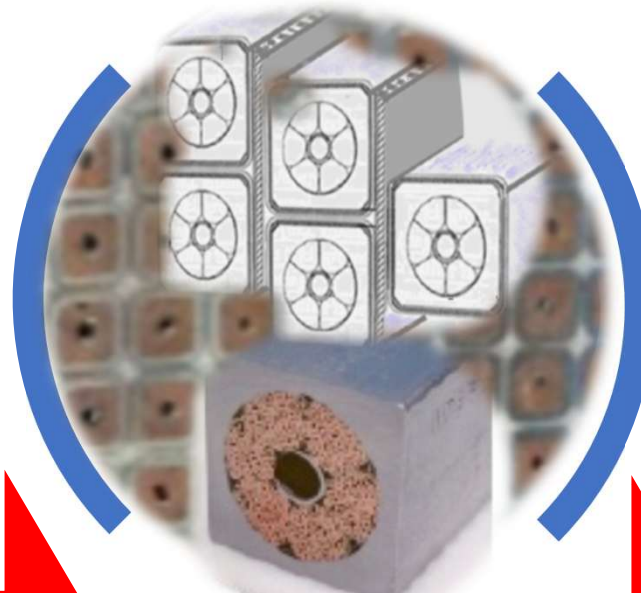
LTS cable and magnets



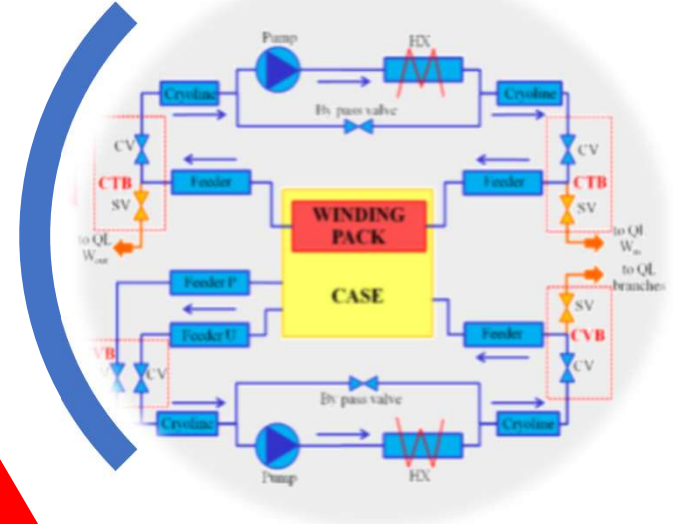
SPATIAL SCALES



MICRO

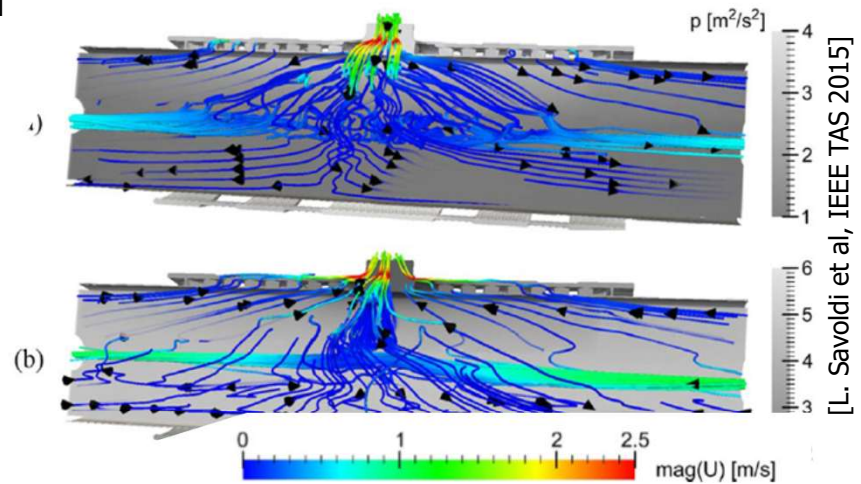
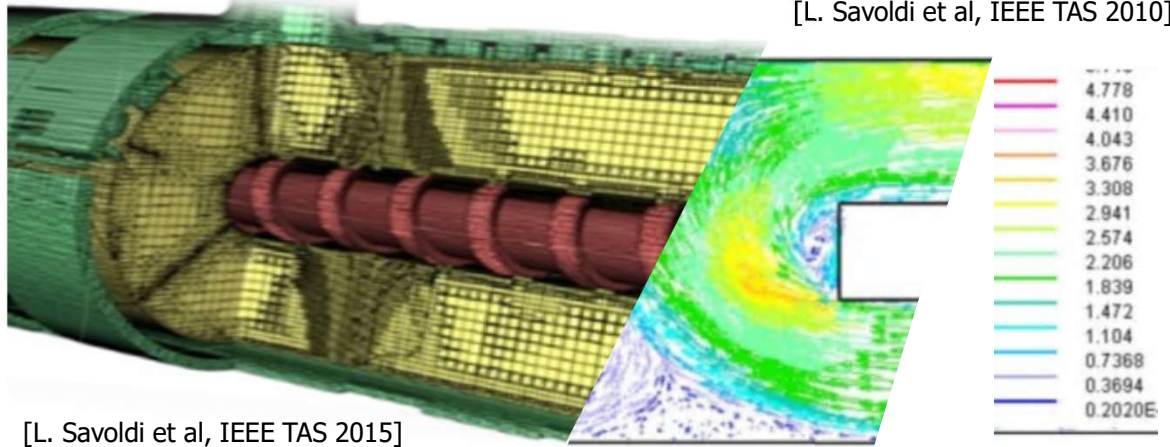


MESO



MACRO

LTS cables at the MICRO scale

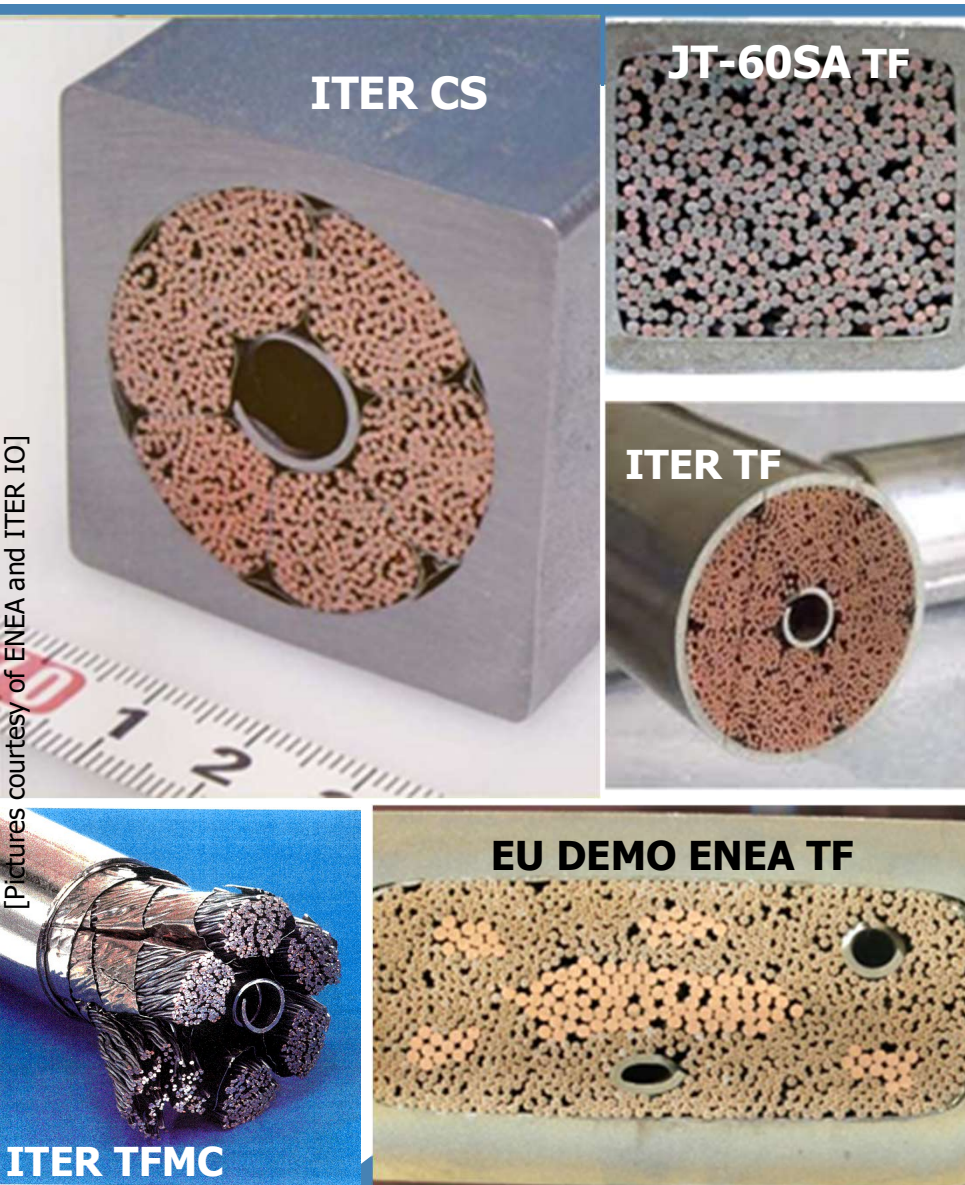


2D/3D CFD as a supplement / backup to experiments

→ develop constitutive relations for 1D models, i.e. correlations/coefficients for

- Major losses (friction factors) in the cable regions
- Minor losses
- Heat transfer coefficients

Experiments to characterize friction / heat transfer /
current distribution in HTS CICC needed to assess model
accuracy (MIT // SPC // KIT // EUROfusion-China)



MESO scale: CICCs

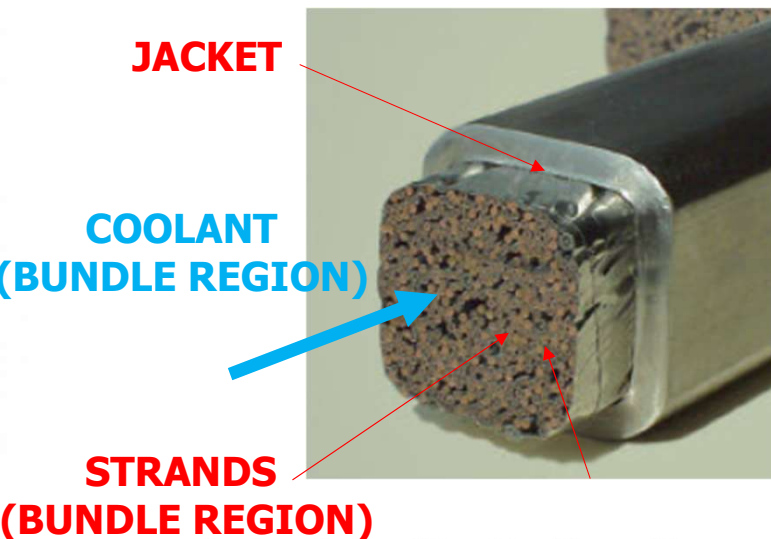


Low critical temperature SC (e.g., Nb_3Sn or NbTi) are manufactured in multi-stage **cable-in-conduit conductors** (CICC) cooled by supercritical He @ ~ 5 K and 0.5 MPa

Physical Architecture: 1-Region vs. 2-Region Configurations

All LTS CICC features a thin or thick jacket contacting both strands and helium, and strands with a massive wetted perimeter twisted in a porous bundle medium.

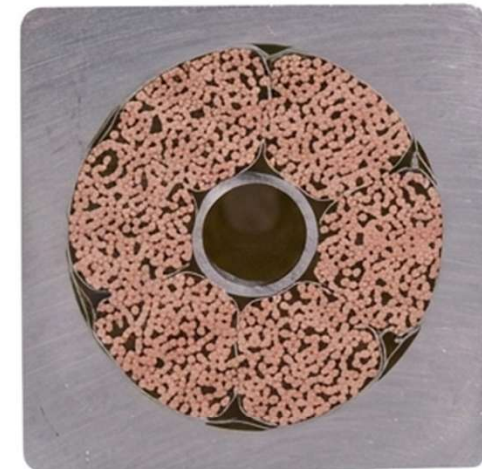
1-Region CICC



A single, unified helium flow region permeating the entire porous strand bundle.

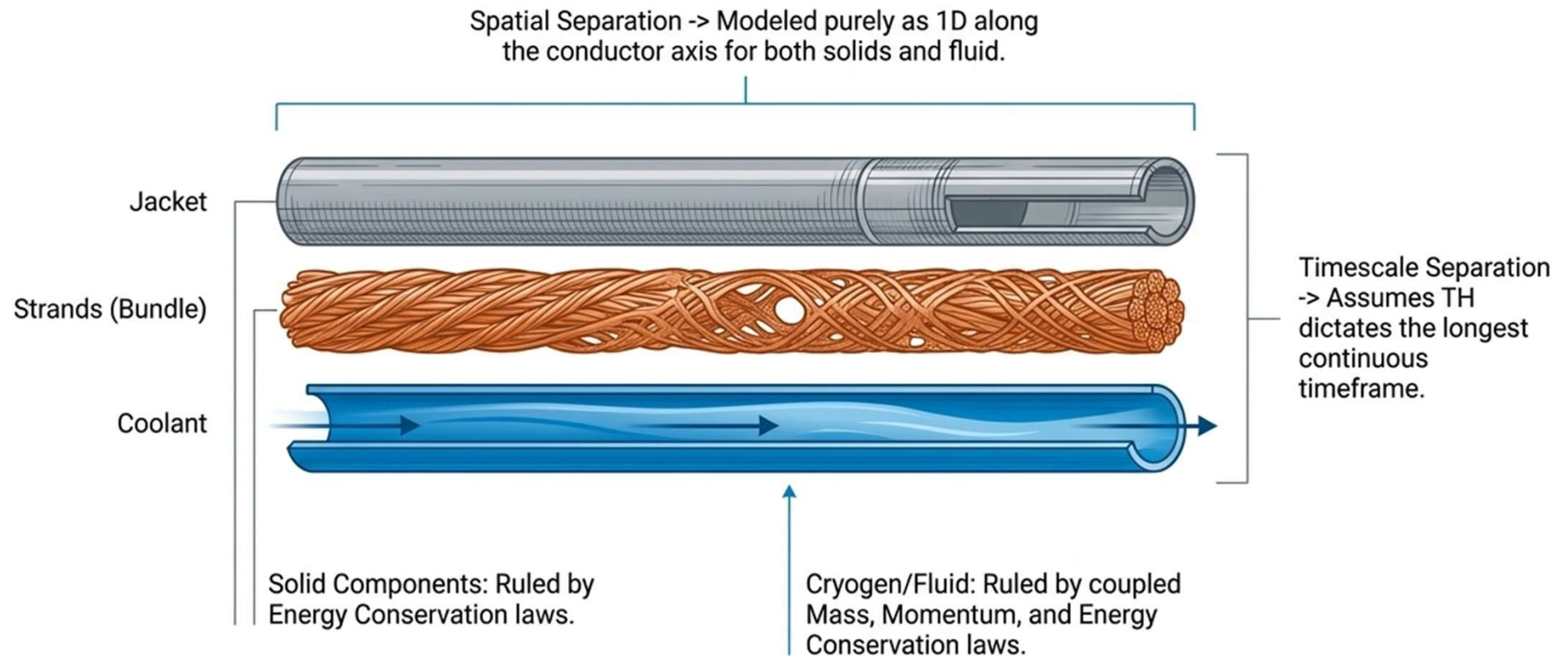
Different layouts may require different model features

2-Regions CICC



Two distinct helium flow regions (an inner channel and the outer bundle) separated by a physically permeable interface.

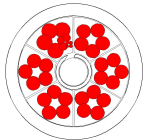
Synthesizing the 1D Thermal-Hydraulic Model



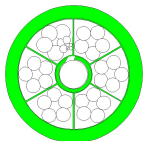
Transient heat conduction in Strands and Jacket



$$A_{st}\rho_{st}c_{p,st}\frac{\partial T_{st}}{\partial t} - A_{st}\frac{\partial}{\partial x}\left(k_{st}\frac{\partial T_{st}}{\partial x}\right) = A'_{st,jk}h_{st,jk}(T_{jk} - T_{st}) + A'_{st,He}h_{st,He}(T_{He} - T_{st}) + Q_J(t, x) + Q_{op}(t, x)$$



$$A_{jk}\rho_{jk}c_{p,jk}\frac{\partial T_{jk}}{\partial t} - A_{jk}\frac{\partial}{\partial x}\left(k_{jk}\frac{\partial T_{jk}}{\partial x}\right) = A'_{st,jk}h_{st,jk}(T_{st} - T_{jk}) + A'_{jk,He}h_{jk,He}(T_{He} - T_{jk}) + Q_{op}(t, x)$$



...each coupled to its initial temperature profile + boundary conditions at the cable inlet/outlet (adiabatic conditions)

Could be also written as thermal resistances

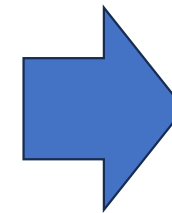
Navier-Stokes equations for 1-region CICC



Mass, momentum and energy conservation in the classical form:

$$\left\{ \begin{array}{l} \frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_j)}{\partial x_j} = 0 \\ \frac{\partial(\rho u)_i}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + S_{v,i} \\ \frac{\partial \rho e}{\partial t} + \frac{\partial(\rho e u_j)}{\partial x_j} = -p \frac{\partial u_j}{\partial x_j} + \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + \Phi + S_e \end{array} \right.$$

$$\tau = \begin{bmatrix} 2\mu \frac{\partial u}{\partial x} - \frac{2}{3}\mu \nabla \cdot \vec{u} & \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \\ \mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) & 2\mu \frac{\partial v}{\partial y} - \frac{2}{3}\mu \nabla \cdot \vec{u} & \mu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \\ \mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) & \mu \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right) & 2\mu \frac{\partial w}{\partial z} - \frac{2}{3}\mu \nabla \cdot \vec{u} \end{bmatrix}$$



**Move to 1D
formulation for
 $Re \gg 1$
→ inviscid flow
(Euler equations)**

Euler-like set of equations for He in 1-region CICCs



$$\begin{aligned}
 (1) \quad & \frac{\partial \rho}{\partial t} + \frac{\partial \rho v}{\partial x} = 0 \quad - 2f \frac{v|v|}{D_h} \\
 (2) \quad & \frac{\partial \rho v}{\partial t} + \frac{\partial \rho v^2}{\partial x} + \frac{\partial p}{\partial x} = \rho \mathbf{F} \\
 (3) \quad & \frac{\partial \rho e}{\partial t} + \frac{\partial \rho e v}{\partial x} + \frac{\partial p v}{\partial x} = \rho v F + \sum h P / A' \Delta T
 \end{aligned}$$

$\Lambda_v =$ the effect of friction is now hidden in the source term for the momentum law

$\Lambda_e =$ Work done by friction forces + thermal coupling to the solids

The variables ρ and e are not directly measurable $\rightarrow$ rewrite the equations using another set of thermodynamic independent variables

From conservative to non-conservative variables



- ❖ Different options available: (ρ, T) , (ρ, i) , (p, T)
- ❖ What should we choose? (think of what we can actually measure...)
- ❖ *Try to re-write the conservation laws in terms of the non-conservative variables: v, p, T*

Derive a new set of Euler-like equations



1. Develop the momentum equation (2) derivatives, multiply (1) by velocity and substitute (2) in (1)
2. Consider the specific energy $e = i + \frac{v^2}{2}$, substitute in (3), develop all the derivatives and collect terms
3. Consider the specific enthalpy $w = i + \frac{p}{\rho} \rightarrow i = w - \frac{p}{\rho}$, replace and develop the products
4. Introduce the Gruneisen parameter $\phi = \left(\frac{1}{\rho} \frac{\partial p}{\partial i} \right)_{\rho} = \left(\frac{\rho}{T} \frac{\partial T}{\partial \rho} \right)_s$ and

$$d\rho = \frac{1+\phi}{c^2} dp - \frac{\rho\phi}{c^2} dw$$
5. Rewrite $di = \left(\frac{p}{\rho} - \phi c_v T \right) \frac{d\rho}{\rho} + c_v dT$ and substitute it in (3)



1° final equation



2° final equation



3° final equation

Euler-like set of equations for the cryogen in 1-region CICC: the Gandalf model



❖ For He in fusion (and power) cables

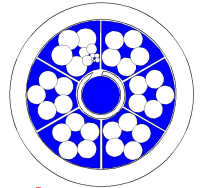
$$\begin{cases} \frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{1}{\rho} \Lambda_v \\ \frac{\partial p}{\partial t} + \rho c^2 \frac{\partial v}{\partial x} + v \frac{\partial p}{\partial x} = \phi (\Lambda_e - v \Lambda_v) \\ \frac{\partial T}{\partial t} + \phi T \frac{\partial v}{\partial x} + v \frac{\partial T}{\partial x} = \frac{1}{\rho c_v} (\Lambda_e - v \Lambda_v) \end{cases}$$

$$\phi = V \left(\frac{dp}{dE} \right)_v = \frac{\alpha K_T V}{c_v}$$

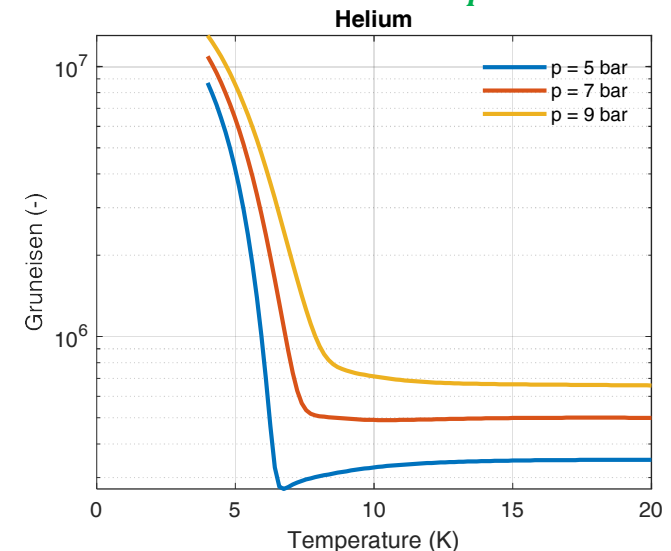
$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$$

Gruneisen parameter

$$K_T = -V \left(\frac{\partial T}{\partial V} \right)_T$$



- RHS **sources/sinks** ($\leftarrow$ interaction with solids and other channels) include **constitutive relations** which require transport coefficients (**friction factors, heat transfer coefficients**) $\leftarrow$ Local 3D models (microscale!)

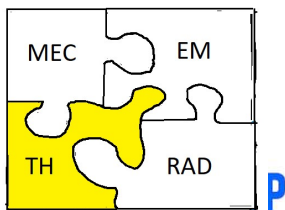


[L. Bottura, J. Comp. Phys., 1996]

Boundary conditions



- ❖ Solid components: adiabatic boundaries
- ❖ Fluid flow:
 - ❖ Usually subsonic (Mach number ~ 0.1 during a quench, well below it in normal operation)
 - ❖ Boundary conditions from linearized characteristic analysis gives:
 - ❖ **INFLOW: prescribe 2 Riemann variables**
 - ❖ **OUTFLOW prescribe 1 Riemann variable**
 - ❖ Typically:
 - ❖ At **inlet**: impose **pressure** and **temperature** (2 variables).
 - ❖ At **outlet**: impose only **pressure** (1 variable).



[L. Bottura, J. Comp. Phys., 1996]

Moving to 2 (or plus) -regions CICCs -I



- ❖ The central channel is a low-impedance region:

$$v_H \gg v_B$$

- ❖ The central channel is not directly in contact with heated solids:

$$T_H \neq T_B$$

- ❖ Two different sets of Euler equations are needed for B and H

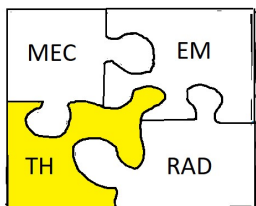
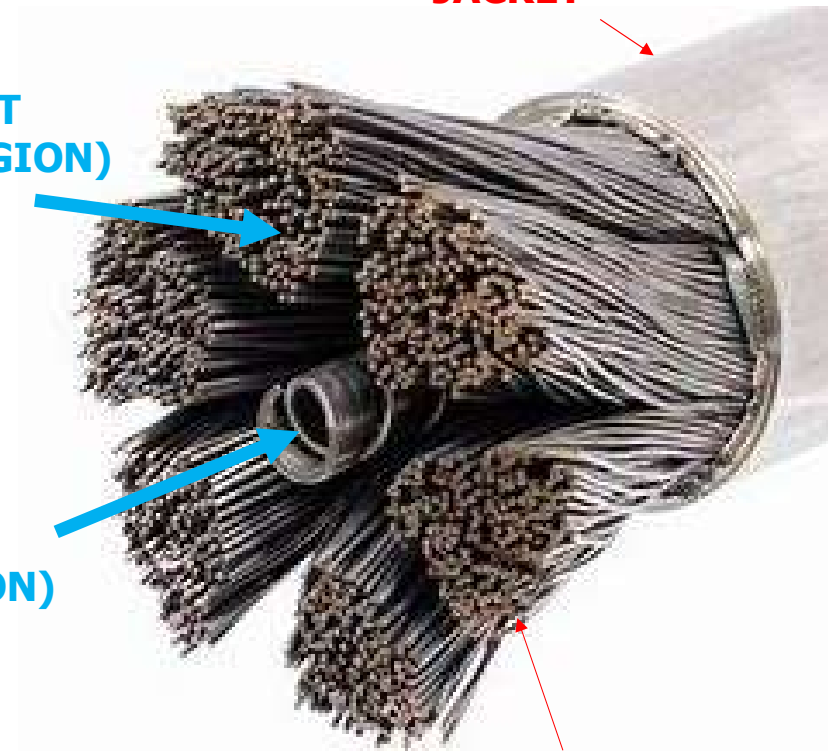
- ❖ The B/H interface is not impermeable

COOLANT
(BUNDLE REGION)

COOLANT
(HOLE REGION)

JACKET

STRANDS
(BUNDLE REGION)



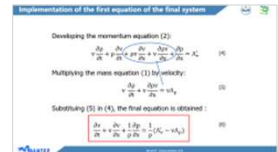
P

Moving to 2 (+)regions CICCs - Mithrandir



- ❖ A source of mass (transporting momentum and energy) is also present to account for the transfer from (to) the Bundle B to (from) the Hole H regions:

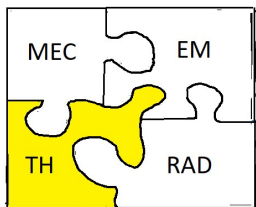
$$\begin{cases} \frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{1}{\rho} [\Lambda'_v - v \Lambda'_\rho] \\ \frac{\partial p}{\partial t} + \rho c^2 \frac{\partial v}{\partial x} + v \frac{\partial p}{\partial x} = \phi \left[\Lambda'_e - v \Lambda'_v - \left(w - \frac{v^2}{2} - \frac{c^2}{\phi} \right) \Lambda'_\rho \right] \\ \frac{\partial T}{\partial t} + \phi T \frac{\partial v}{\partial x} + v \frac{\partial T}{\partial x} = \frac{1}{\rho c_v} \left[\Lambda'_e - v \Lambda'_v - \left(w - \frac{v^2}{2} - \phi c_v T \right) \Lambda'_\rho \right] \end{cases}$$



How can the mass-momentum-energy transfer among the B and H region be described? Pseudo-valve model

$$\Lambda_\rho = \frac{K(p_B - p_H)}{lA}$$

[R. Zanino et al, Journal of Fusion Energy, 1995]



OPENSC²: a novel object-oriented tool

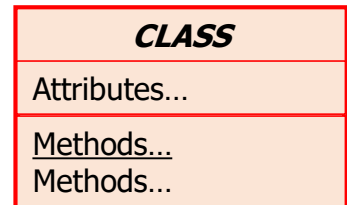
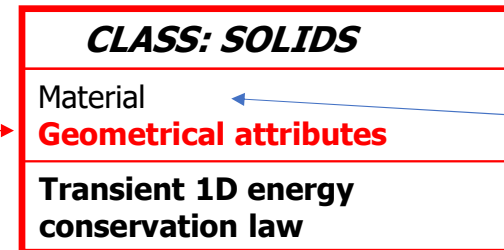
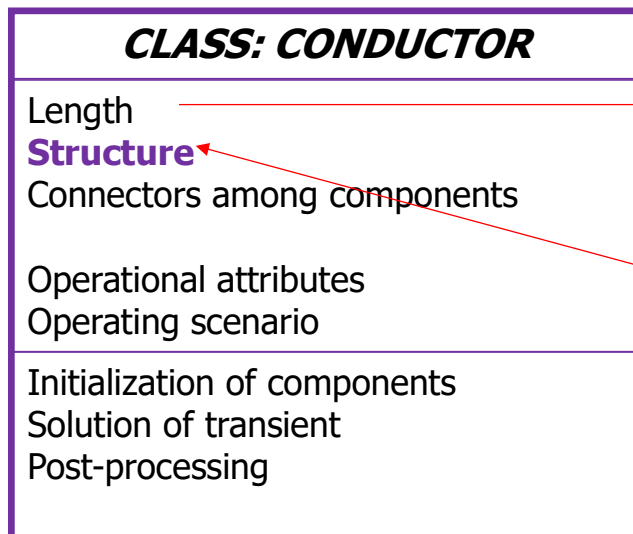


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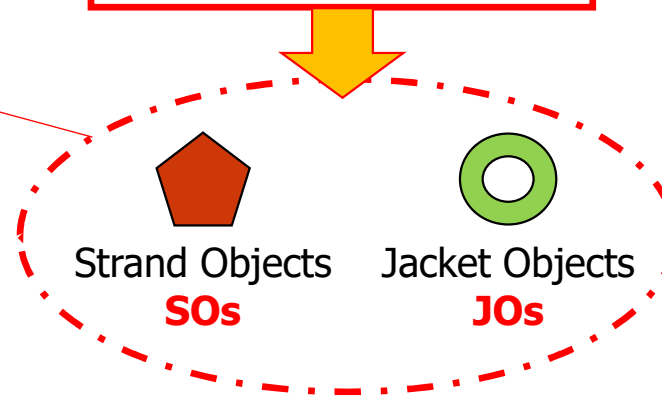


Object-oriented approach implemented in Python

*Each conductor object **CO**
from class CONDUCTOR*



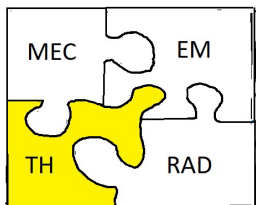
**LIBRARY
SOLID MEDIA
(open database)**



***SOs and JOs are
"children"(instances)
of the class SOLIDS***

[L. Savoldi et al, Cryogenics, 2022]

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OPENSC²: a novel object-oriented tool

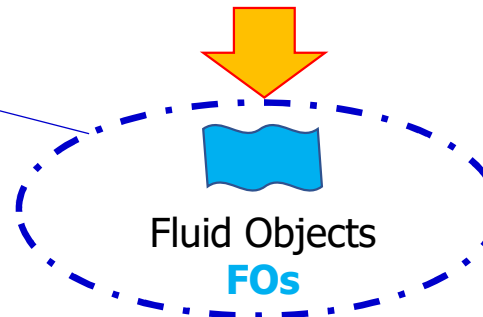
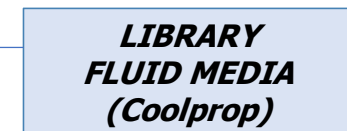
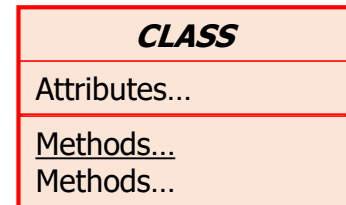
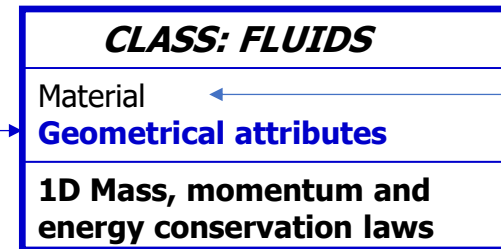
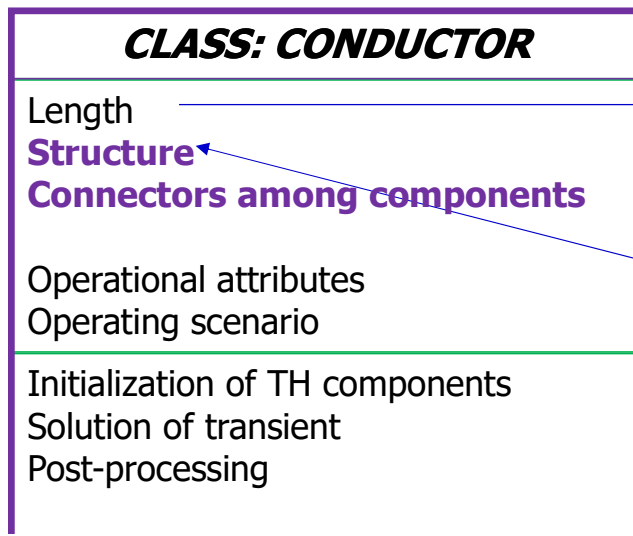


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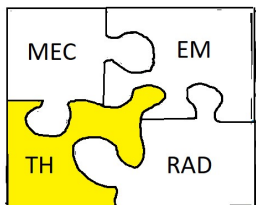
Object-oriented approach implemented in Python

*Each conductor object **CO**
from class CONDUCTOR*



*FOs are instances of
the class FLUIDS*

*The tool is intrinsically multi-conductor: several **COs** can be instantiated simultaneously*



OPENSC²: connectors between objects

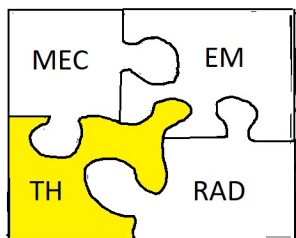


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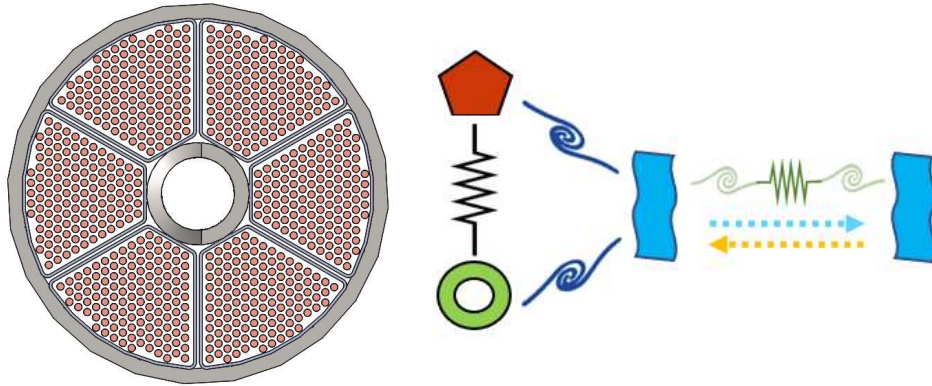


→ Each **CO** is assembled using **SOs**, **FOs** and **Jos**, connected through connectors

Object	Interaction	Symbol	Object
SO 	Thermal by conduction		 SO
SO 	Thermal by conduction		 JO
JO 	Thermal by conduction		 JO
JO 	Thermal by radiation		 JO
SO 	Thermal by convection		 JO
JO 	Thermal by convection		 FO
FO 	Thermal through an impermeable surface		 FO
FO 	Hydraulic and thermal by a permeable surface		 FO



Model of an ITER TF conductor



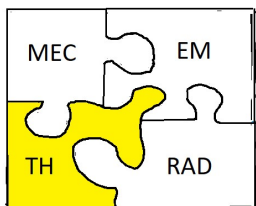
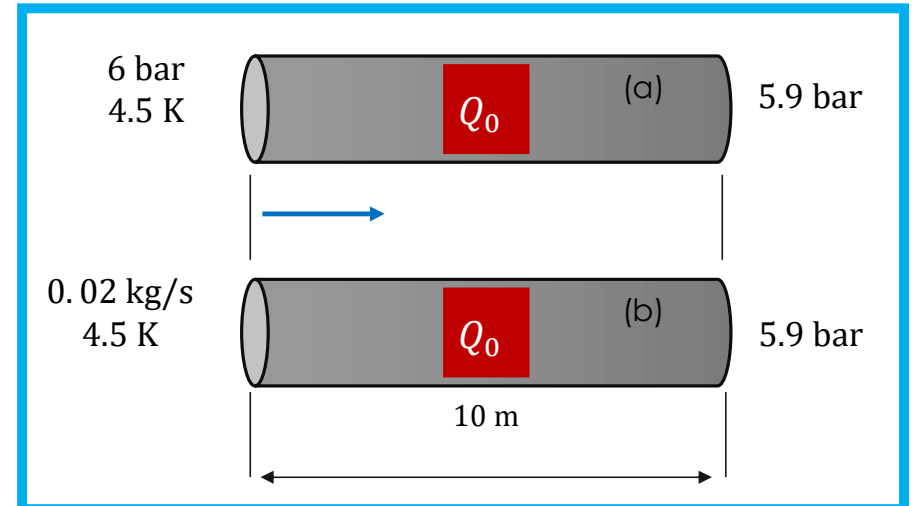
Heat the central 2 m of the strands with 250 W/m for 10 s .

Different sets of BCs

Instance of the Conductor class assembled as:

$$\text{CO} = 1 \text{ SO} + 1 \text{ JO} + 2 \text{ FOs}$$

Consider a 10m-long conductor



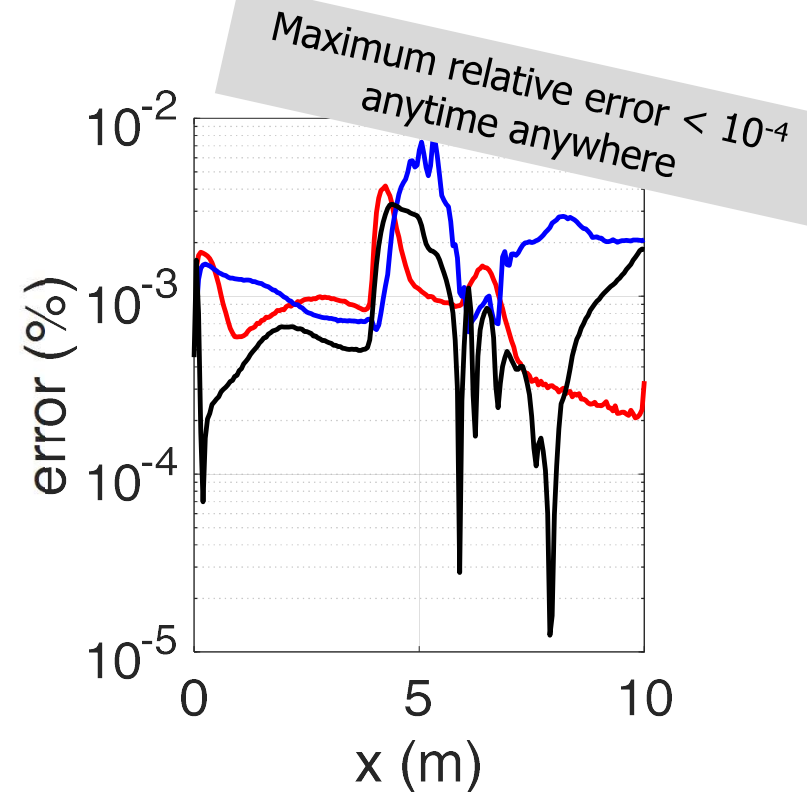
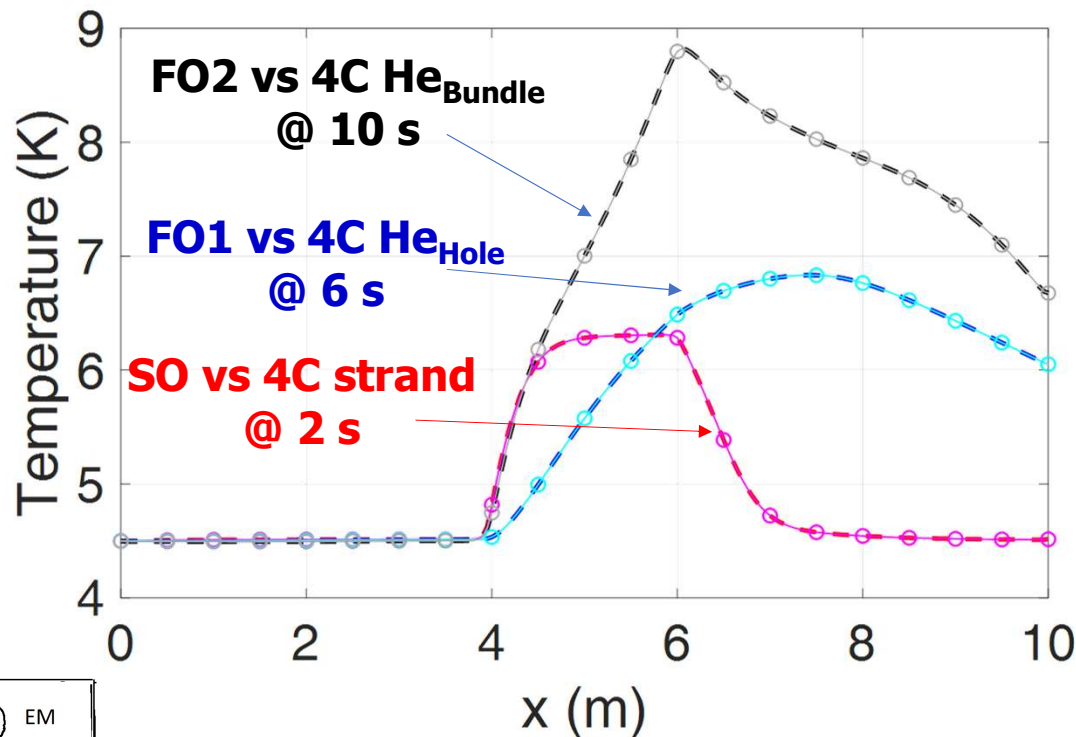
ITER TF conductor: code-to-code benchmark



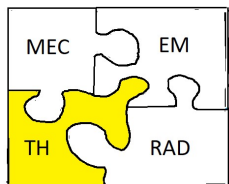
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- Test @ **Constant inlet velocity**, space and time convergence checked (1st order here)
- BENCHMARK with the Mithrandir code



- Similar pictures for all sets of boundary conditions / drivers tested so far



TEP

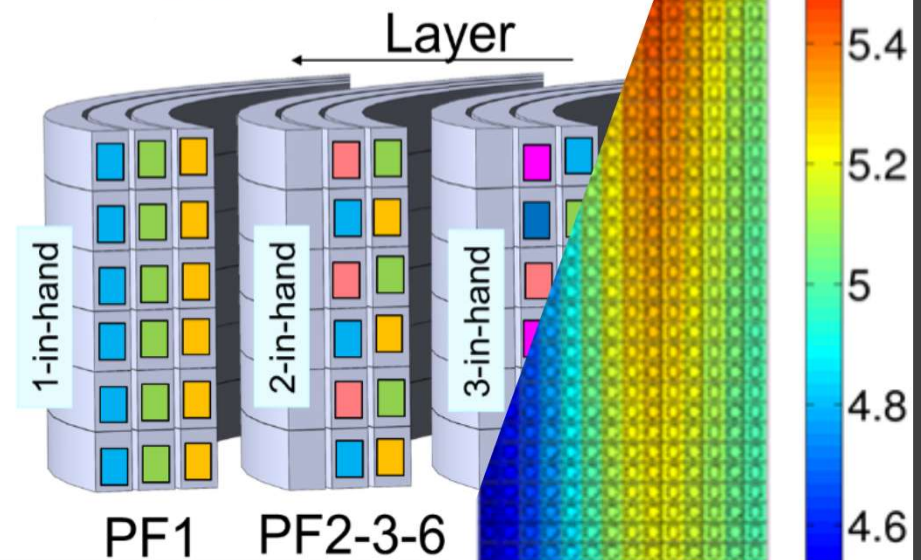
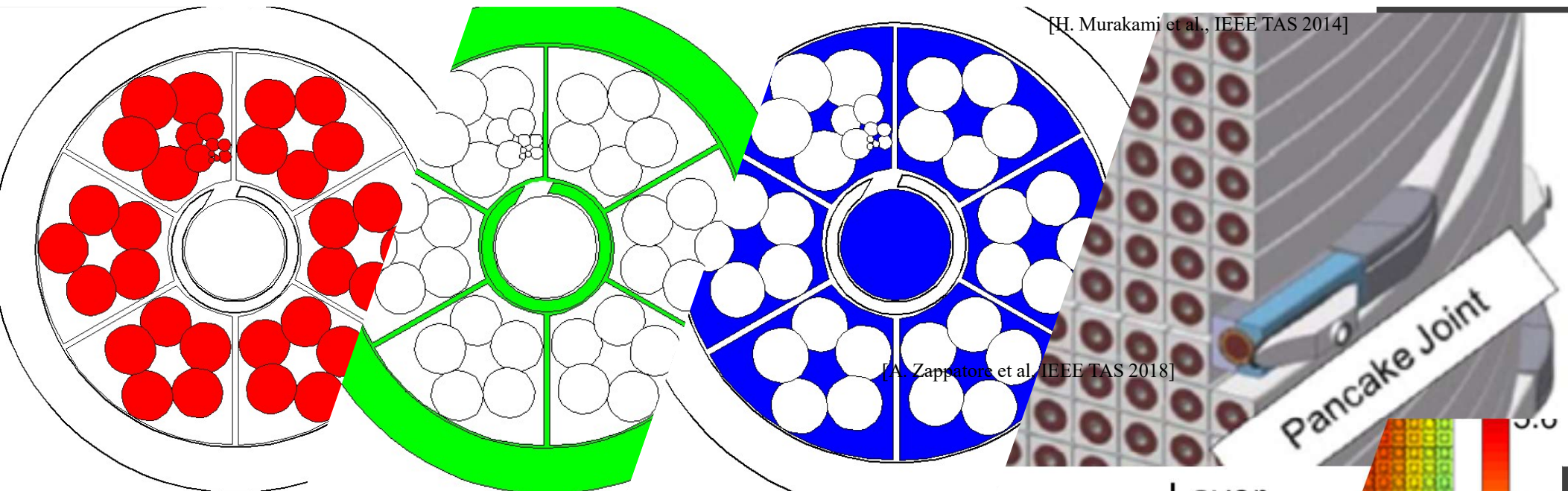
LTS modelling: meso-scale

- Space scale separation allows 1D (axial) transient models for any CICC: He in the hole + He in the bundle + SC strands + Jacket
- Time scale separation along and across CICC → solve 3D problem as several coupled 1D problems.



Winding model

NEA JRC, Spring semester 2026



Multi-conductor Mithrandir (M&M) code

[L. Savoldi et al, 2001]

Time scale separation along and across CICC → solve 3D problem as several coupled 1D problems.

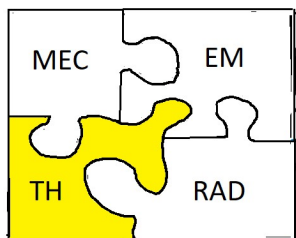
Use Mithrandir model for each CICC

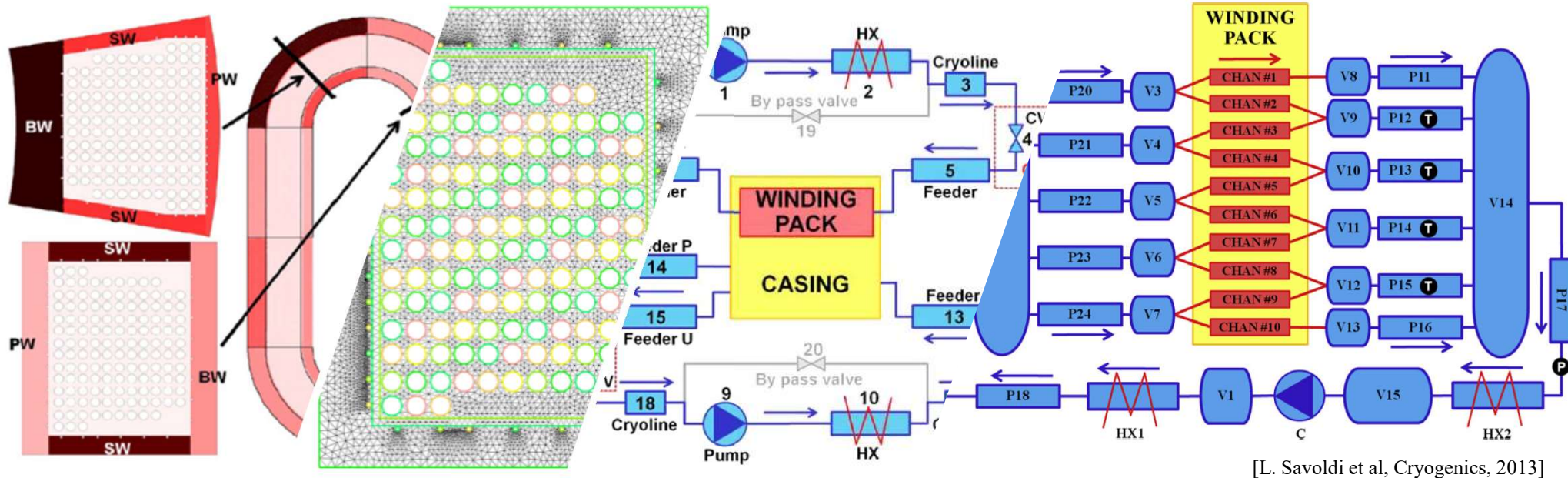
Transverse coupling is explicit in time

Longitudinal coupling with circuit code

Very general coil topology can be simulated with this strategy!

[L. Savoldi et al, Cryogenics, 2000]

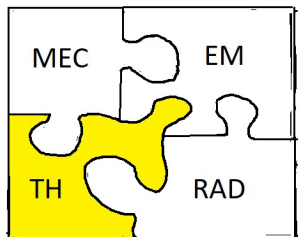




- Poloidal discretization of 3D transient heat conduction problem using several *2D cuts on suitable cross sections* → solve transient 2D heat conduction with FE

- Cryogenic circuit 0D/1D models: Pumps, valves, HX, cryolines, LHe bath, ...using MODELICA

[L. Savoldi Richard et al, Cryogenics, 2010]



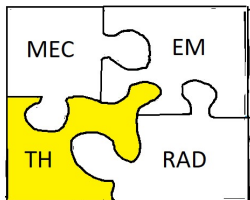
Thermal-hydraulic behavior of LTS cables: fast transients

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Dynamic of stability transients (I)

- *How fast is a stability "fast" transient? Time-scale?*
- *What do we need for modelling?*
 - *Transient heat transfer coefficient evaluation for thermal boundary layers*
 - *Good estimation of HTC in general*
 - *Robust numerics (very small time stepping required)*

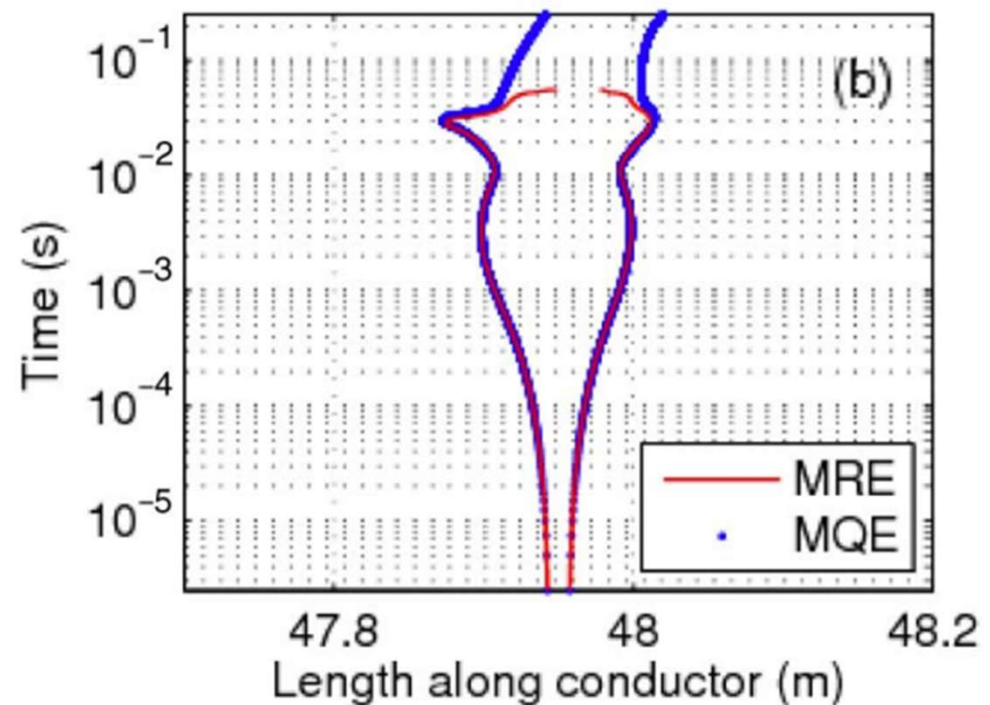
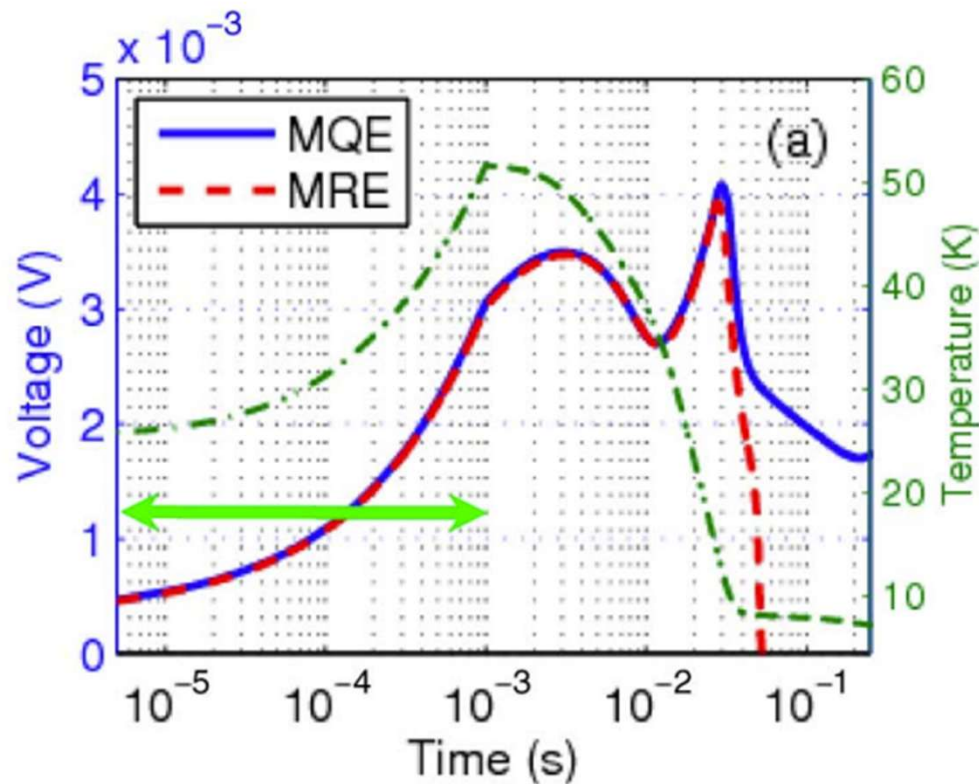
AIM = determine the MQE (Minimum Quench Energy) and the MRE (Maximum Recovery Energy)



Thermal-hydraulic behavior of LTS cables: fast transients

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Dynamic of stability transientsn(II)

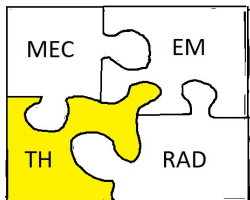


Thermal-hydraulic behavior of LTS cables: fast transients

Dynamic of quench evolution (I)

- *How fast is a quench "fast" transient? Time-scale?*
- *What do we need for modelling?*
 - *In 2-regiosn CICC, very accurate description of the mass and heat transfer among B and H*
 - *Good estimation of HTC in general*
 - *Realistic boundary conditions (circuit model?)*
 - *Robust numerics (adaptive time stepping)*

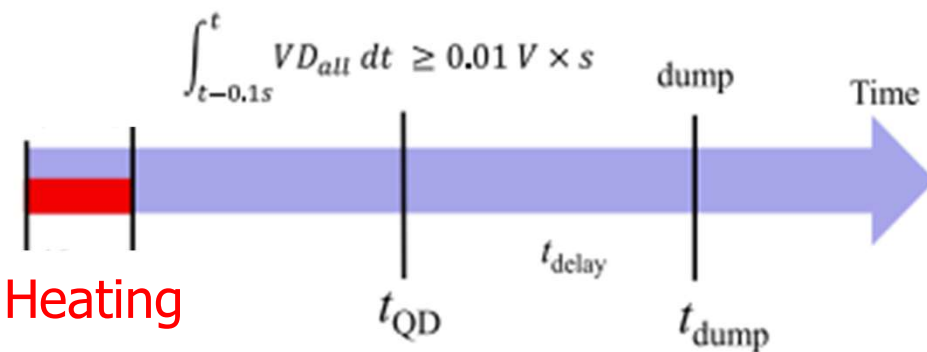
AIM = determine the hot spot temperature and peak pressure + quench evolution speed



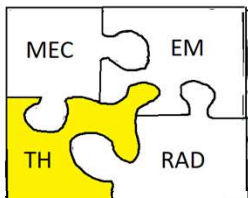
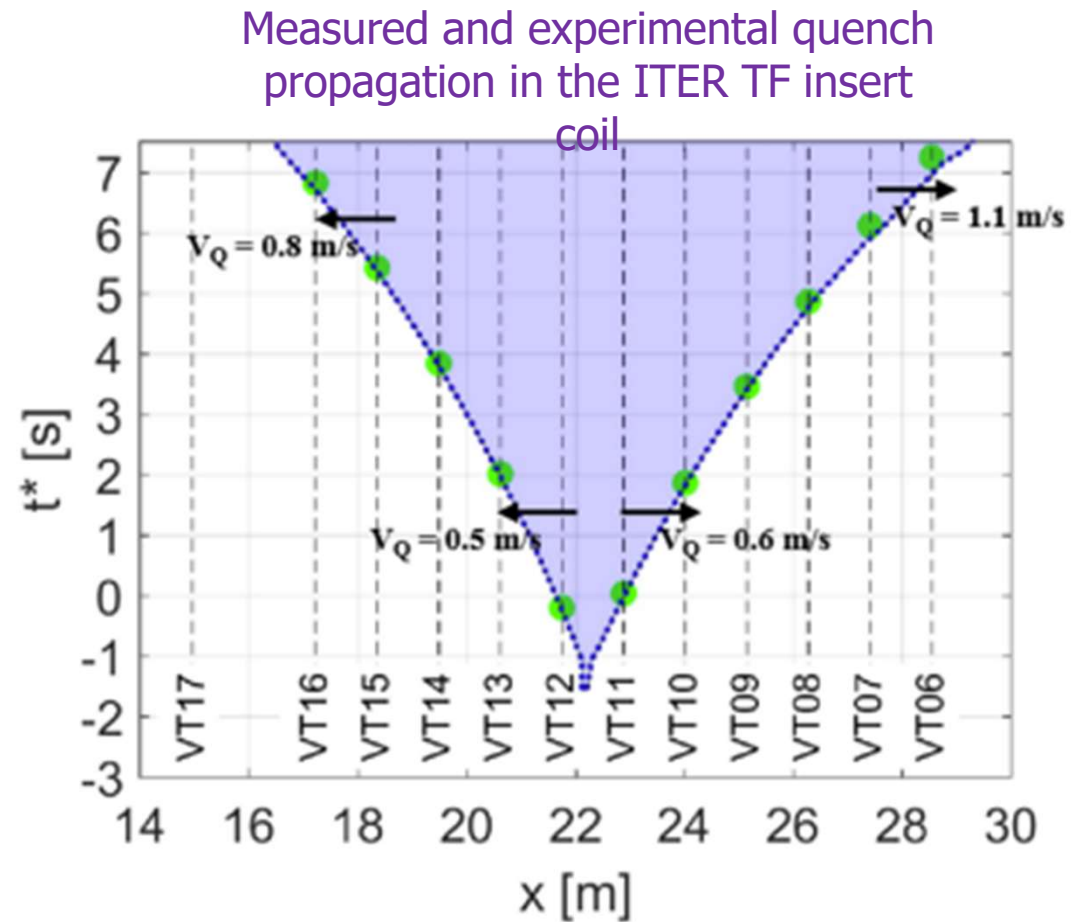
Thermal-hydraulic behavior of LTS cables: fast transients

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Dynamic of quench evolution



Heating



[R. Zanino, ... L. Savoldi, 2018]

NELAHTP - Sprin

Thermal-hydraulic behavior of LTS cables: slow transients

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Evaluation of temperature margin in operation

How slow is a temperature margin transient evaluation?

Evaluation of the temperature margin in operation requires:

- Winding pack geometry
- Magnetic field (x, y, z, t)
- Heat loads (x, y, z, t)
- Transport current (t)
- Coolant boundary conditions (t)
- Strand scaling for SC props

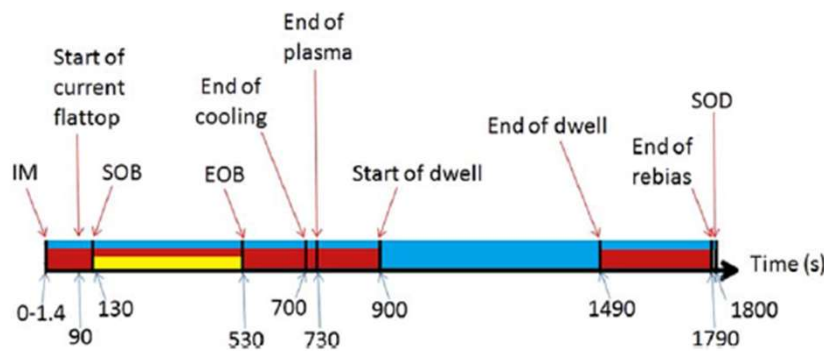


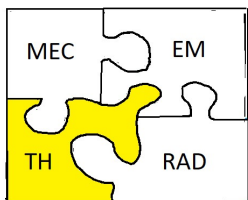
Fig. 2. Time line of the standard 15 MA scenario foreseen in ITER (IM = initial magnetization, SOB = start of burn, EOB = end of burn, SOD = start of discharge). The radiative (static) load is applied during all the 1800 s (blue line), the eddy current load according to the red line, and the nuclear load between the SOB and the EOB (yellow line).

[L. Savoldi Richard et al, IEEE TAS, 2014]

$$\rightarrow \Delta T_{mar}(\xi, t) = T_{cs}(\xi, t) - T_{strand}(\xi, t)$$

$$AIM = \text{determine } \Delta T_{mat}^{min} = \min_{\xi, t} \Delta T_{mar}(\xi, t)$$

NELAHTP - Spring semester 2026

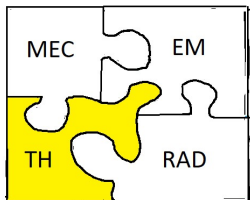


Thermal-hydraulic behavior of LTS magnets

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Dynamic of quench evolution

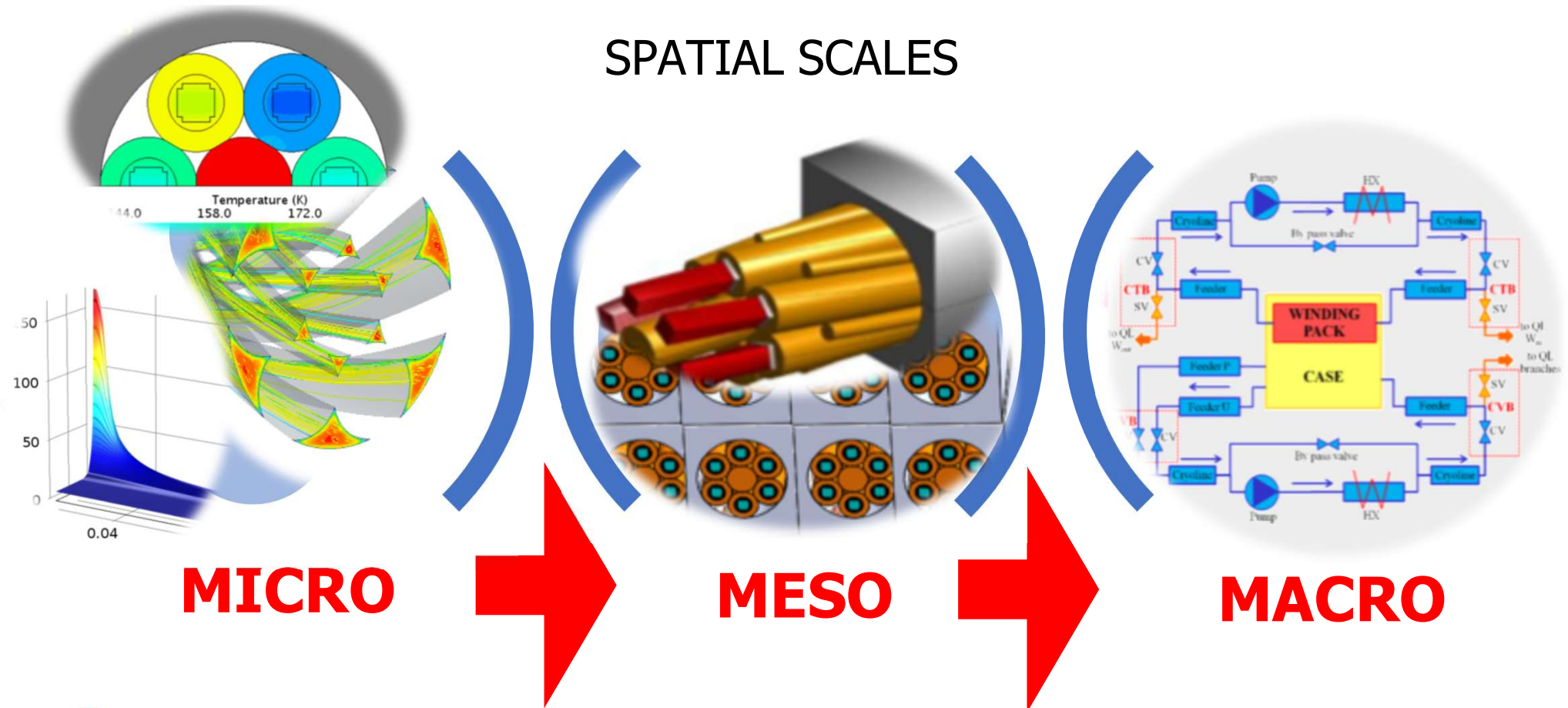
Which are the new effects that should be accounted for when an entire magnet is quenching?



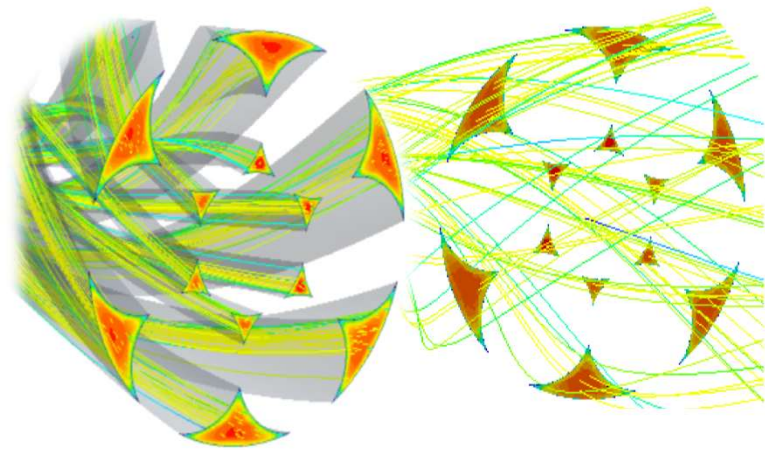
HTS cable and magnets for fusion



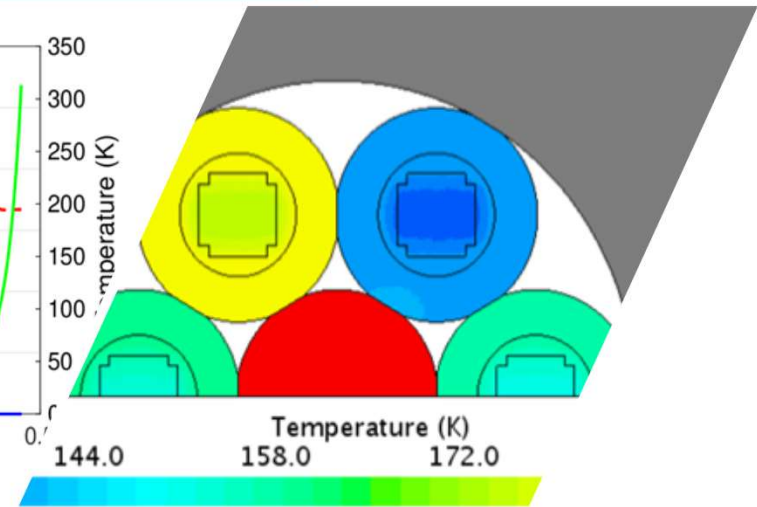
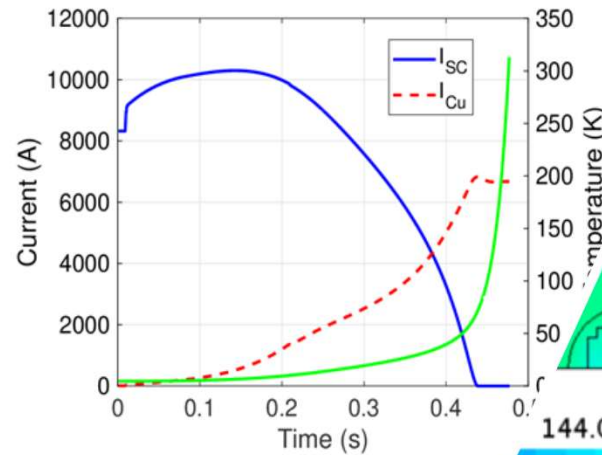
SPATIAL SCALES



HTS cables at the MICRO scale (I)



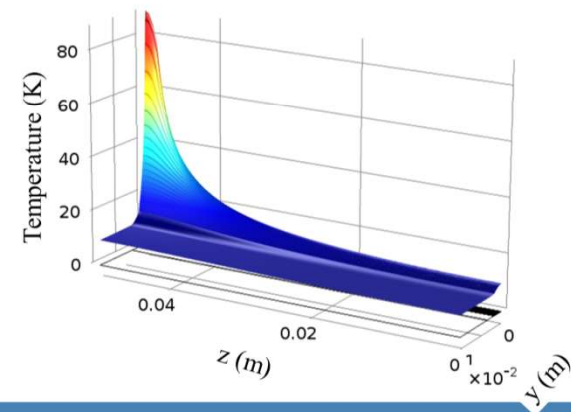
[A. Zappatore, L. Savoldi et. al, SuST, 2018]



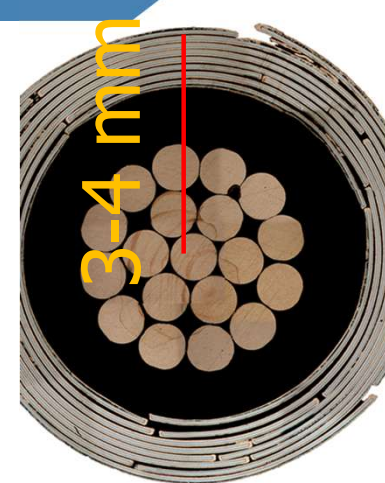
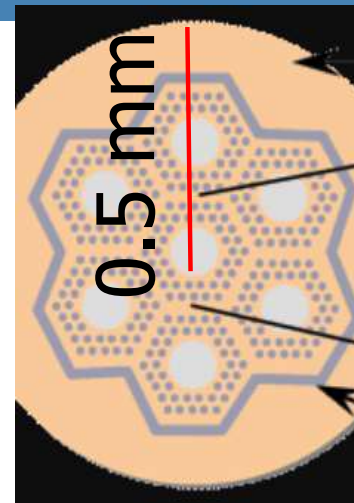
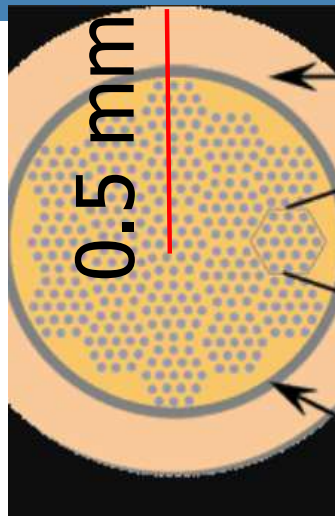
2D/3D CFD as a supplement / backup to experiments

→ develop constitutive relations for 1D models, i.e. correlations/coefficients for

- Major/minor losses in the cable regions
- Thermal coupling among CICC components on the cross section

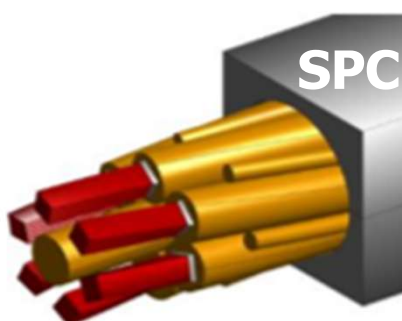
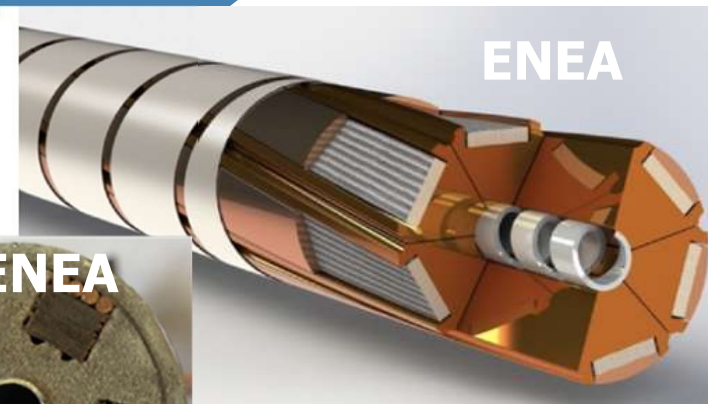
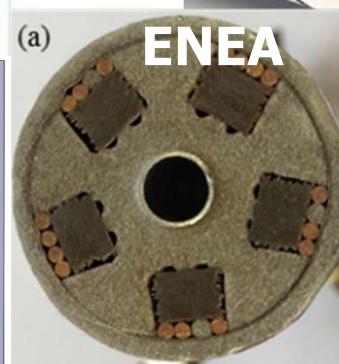
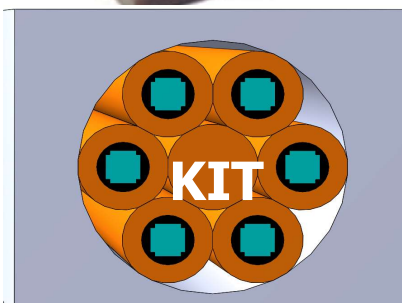
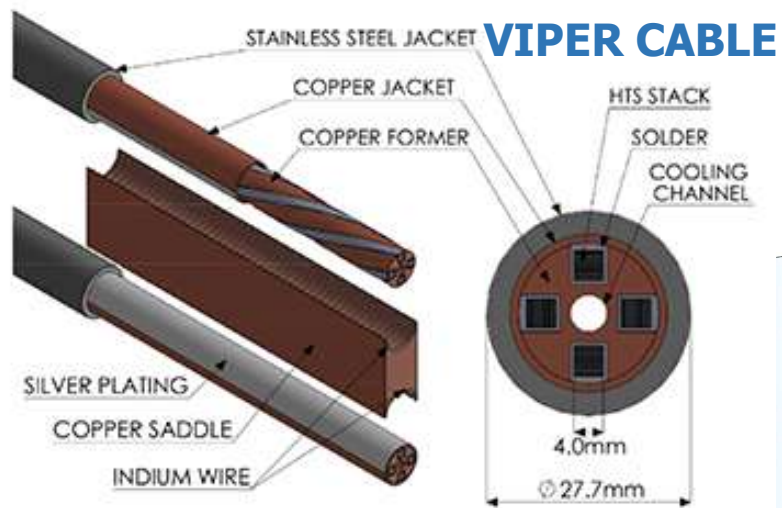


Towards HTS CICC: HTS vs LTS strands



- Different size (wetted perimeter)
- Different material properties (e.g. $k_{\text{HTS}} < k_{\text{LTS}} \rightarrow$ Larger thermal gradients in HTS in case of fast transients)

HTS CICC



Several conductor concepts for fusion magnets based on High Temperature Superconductors (HTS) are being investigated worldwide, most based on CICC concept

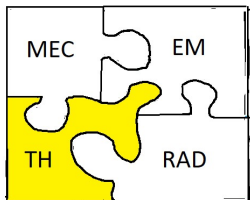
[Pictures courtesy of MIT, NCEPU, ACT, ENEA, SPC, MIT, NIFS]

Thermal-hydraulic behavior of HTS cables

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Peculiarities of HTS CICC's for fusion

*How the model developed for LTS cables
could be modified? New needs?
Challenges?*



Model of an HTS cable for fusion



Politecnico
di Torino

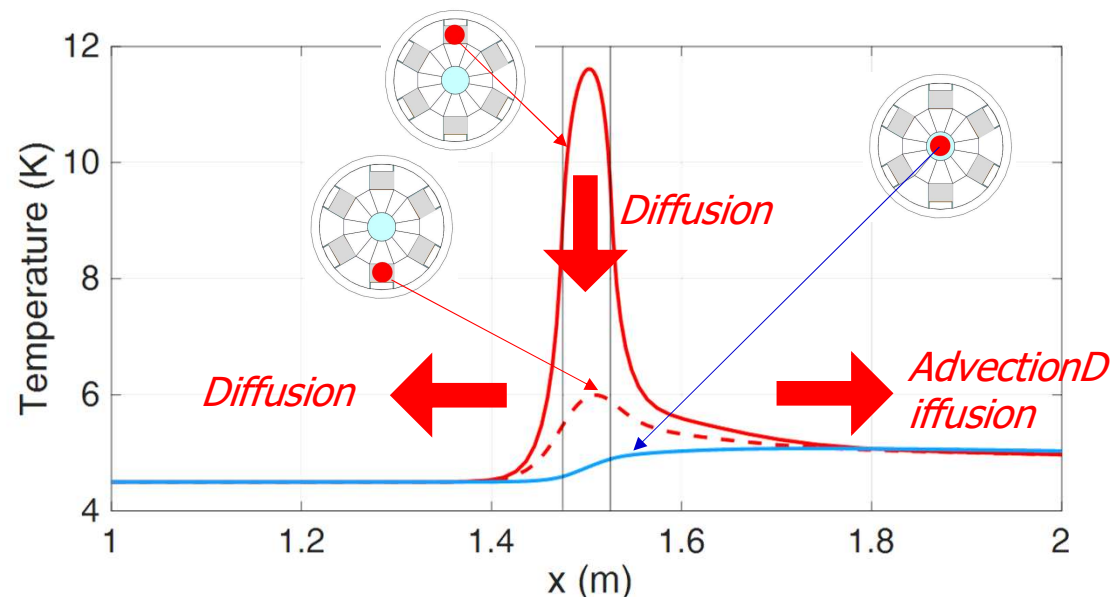
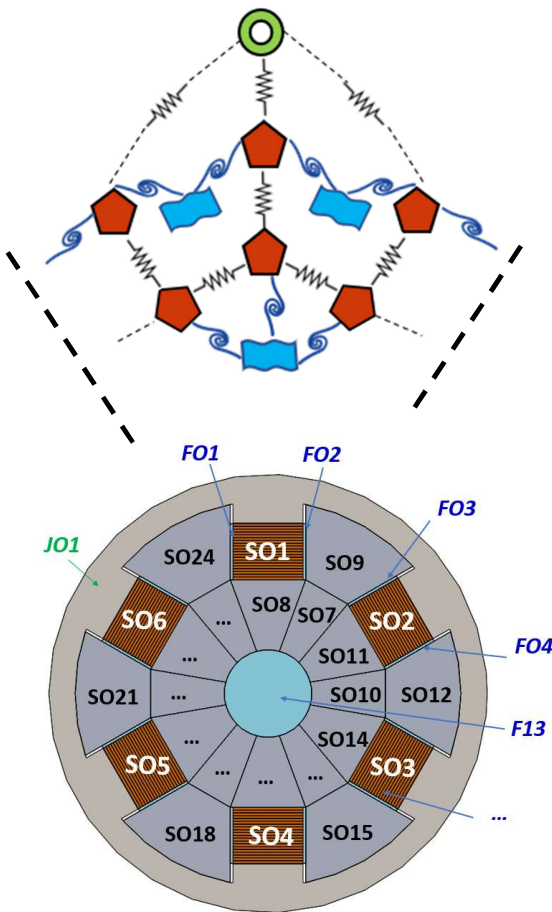


- Instance of the Conductor class assembled as:

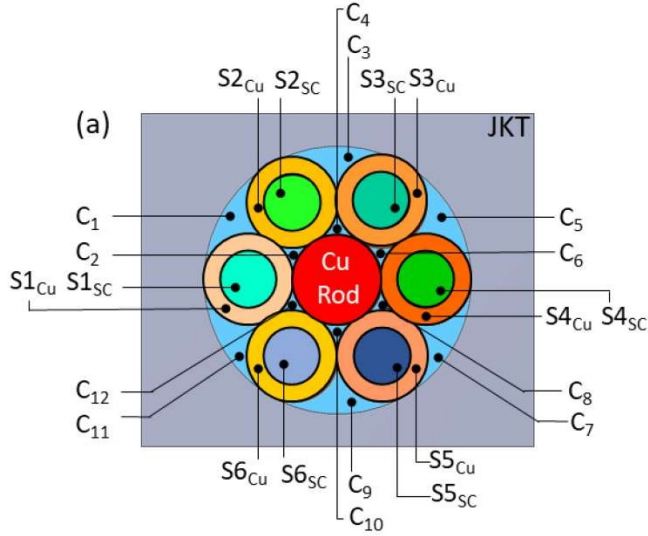
$$\text{CO} = (18+6) \text{ SO} + 1 \text{ JO} + 13 \text{ FOs}$$

[A. Zappatore et al, IEEE TAS 2020]

- Consider a SULTAN-like 3m-long conductor, cooled by 5 g/s of SHe @ 4.5 K, 10 bar; heat the central 5 cm of SO1 with 250 W/m for 1 s



A novel TH-EM HTS cable model



$$\bar{R} = \begin{bmatrix} r_{1,1} & 0 & \dots & 0 \\ 0 & \ddots & & \\ \vdots & & r_{i,i} & \\ 0 & \dots & & \ddots \end{bmatrix}$$

$$\bar{L} = \begin{bmatrix} l_{1,1} & l_{1,2} & \dots & l_{1,n} \\ l_{2,1} & \ddots & & \\ \vdots & & l_{i,i} & \\ l_{n,1} & \dots & & \ddots \end{bmatrix}$$

$$(\bar{G}\bar{L})_k \frac{\partial \bar{I}_k}{\partial t} + \frac{\partial^2 \bar{I}_k}{\partial x^2} + (\bar{G}\bar{R})_k \cdot \bar{I}_k = 0$$

$$\bar{G} = \begin{bmatrix} -\sum_{i=1}^{N_{elec}} g_{1,i} & g_{1,2} & \dots & g_{1,n} \\ g_{2,1} & \ddots & & \\ \vdots & & \ddots & \\ g_{n,1} & \dots & & -\sum_{i=1}^{N_{elec}} g_{n,i} \end{bmatrix}$$

$$\left\{ \begin{array}{l} (\rho c_p A)_{sc,i} \frac{\partial T_{sc,i}}{\partial t} - \frac{\partial}{\partial x} \left[(kA)_{sc,i} \frac{\partial T_{sc,i}}{\partial x} \right] = U_{sc \leftrightarrow Cu,i} \cdot (T_{Cu,i} - T_{sc,i}) + \sum_{j=1}^{N_{fluids}} (Ph)_{sc,i \leftrightarrow j} \cdot (T_j - T_{sc,i}) + q'_{sc,i} \\ (\rho c_p A)_{Cu,i} \frac{\partial T_{Cu,i}}{\partial t} - \frac{\partial}{\partial x} \left[(kA)_{Cu,i} \frac{\partial T_{Cu,i}}{\partial x} \right] = \sum_{p=1}^{N_{macrostrands}} U_{p \leftrightarrow Cu,i} \cdot (T_p - T_{Cu,i}) + \sum_{j=1}^{N_{fluids}} (Ph)_{Cu,i \leftrightarrow j} \cdot (T_j - T_{Cu,i}) + q'_{Cu,i} \end{array} \right.$$

$$\left\{ \begin{array}{l} \frac{\partial v_j}{\partial t} + v_j \frac{\partial v_j}{\partial x} + \frac{1}{\rho_j} \frac{\partial p_j}{\partial x} + 2f_j v_j \frac{|v_j|}{D_{h,j}} = 0 \\ \frac{\partial p_j}{\partial t} + v_j \frac{\partial p_j}{\partial x} + (\rho c^2)_j \frac{\partial v_j}{\partial x} - 2(\phi \rho f v^2)_j \frac{|v_j|}{D_{h,j}} = \frac{\phi_j}{A_j} \left(\sum_{i=1}^{N_{macrostrands}} (Ph)_{sc,i \leftrightarrow j} \cdot (T_{sc,i} - T_j) + \sum_{i=1}^{N_{macrostrands}} (Ph)_{Cu,i \leftrightarrow j} \cdot (T_{Cu,i} - T_j) + q'_j \right) \\ \frac{\partial T_j}{\partial t} + v_j \frac{\partial T_j}{\partial x} + \phi_j T_j \frac{\partial v}{\partial x} - 2(f v^2)_j \frac{|v_j|}{(c_v D_h)_j} = \frac{1}{(c_v \rho A)_j} \left(\sum_{i=1}^{N_{macrostrands}} (Ph)_{sc,i \leftrightarrow j} \cdot (T_{sc,i} - T_j) + \sum_{i=1}^{N_{macrostrands}} (Ph)_{Cu,i \leftrightarrow j} \cdot (T_{Cu,i} - T_j) + \phi_j q'_j \right) \end{array} \right.$$

[A. Zappatore, L. Savoldi et al, SuST 2020]

Implementation of the first equation of the final system



Developing the momentum equation (2):

$$v \frac{\partial \rho}{\partial t} + \rho \frac{\partial v}{\partial t} + \rho v \frac{\partial v}{\partial x} + v \frac{\partial \rho v}{\partial x} + \frac{\partial p}{\partial x} = \Lambda'_v \quad (4)$$

Multiplying the mass equation (1) by velocity:

$$v \frac{\partial \rho}{\partial t} + v \frac{\partial \rho v}{\partial x} = v \Lambda_\rho \quad (5)$$

Substituting (5) in (4), the final equation is obtained :

$$\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{1}{\rho} (\Lambda'_v - v \Lambda_\rho) \quad (6)$$

Implementation of the second equation of the final system (1)



Considering the specific energy as:

$$e = i + \frac{v^2}{2}$$

Where:

- i is the internal specific energy
- $\frac{v^2}{2}$ is the kinetic specific energy

Separating the sum in the energy conservation (3) and collecting terms:

$$\frac{\partial \rho i}{\partial t} + \frac{v^2}{2} \left(\frac{\partial \rho}{\partial t} + \frac{\partial \rho v}{\partial x} \right) + \rho v \frac{\partial v}{\partial t} + \frac{\partial \rho i v}{\partial x} + \rho v^2 \frac{\partial v}{\partial x} + \frac{\partial p v}{\partial x} = \Lambda'_e \quad (7)$$

(Note: A blue circle is drawn around the term $\frac{v^2}{2} \left(\frac{\partial \rho}{\partial t} + \frac{\partial \rho v}{\partial x} \right)$ with an arrow pointing to Λ_ρ)

Developing all derivatives and collecting terms:

$$\rho \frac{\partial i}{\partial t} + i \left(\frac{\partial \rho}{\partial t} + \frac{\partial \rho v}{\partial x} \right) + \rho v \left(\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} \right) + p \frac{\partial v}{\partial x} + \rho v \frac{\partial i}{\partial x} = \Lambda'_e - \frac{v^2}{2} \Lambda_\rho \quad (8)$$

(Note: A blue circle is drawn around the term $i \left(\frac{\partial \rho}{\partial t} + \frac{\partial \rho v}{\partial x} \right)$ with an arrow pointing to Λ_ρ . Another blue circle is drawn around the term $\rho v \left(\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} \right)$ with an arrow pointing to $\frac{1}{\rho} (\Lambda'_v - v \Lambda_\rho)$)

Implementation of the second equation of the final system (2)



Managing (8), equation (9) is obtained:

$$\rho \frac{\partial i}{\partial t} + \rho v \frac{\partial i}{\partial x} + p \frac{\partial v}{\partial x} = \Lambda_e - v \Lambda'_v - \left(i - \frac{v^2}{2} \right) \Lambda_\rho \quad (9)$$

Introducing the specific enthalpy:

$$w = i + \frac{p}{\rho} \rightarrow i = w - \frac{p}{\rho}$$

Replacing and managing (9):

$$\frac{\partial w}{\partial t} + v \frac{\partial w}{\partial x} - \frac{\partial}{\partial t} \left(\frac{p}{\rho} \right) - v \frac{\partial}{\partial x} \left(\frac{p}{\rho} \right) + \frac{p}{\rho} \frac{\partial v}{\partial x} = \frac{1}{\rho} \left[\Lambda'_e - v \Lambda'_v - \left(i - \frac{v^2}{2} \right) \Lambda_\rho \right] \quad (10)$$

Developing the
derivatives and
collecting terms

$$\rightarrow -\frac{1}{\rho} \left(\frac{\partial p}{\partial t} + v \frac{\partial p}{\partial x} \right) + \frac{p}{\rho^2} \left(\frac{\partial \rho}{\partial t} + v \frac{\partial \rho}{\partial x} \right) \quad (11)$$

Implementation of the second equation of the final system (3)



Substituting (11) in (10):

$$\begin{aligned} \frac{\partial w}{\partial t} + v \frac{\partial w}{\partial x} - \frac{1}{\rho} \left(\frac{\partial p}{\partial t} + v \frac{\partial p}{\partial x} \right) + \frac{p}{\rho^2} \left(\frac{\partial \rho}{\partial t} + v \frac{\partial \rho}{\partial x} + \rho \frac{\partial v}{\partial x} \right) &= \frac{1}{\rho} \left[\Lambda'_e - v \Lambda'_v - \left(i - \frac{v^2}{2} \right) \Lambda_\rho \right] \end{aligned} \quad (12)$$


Recalling that: $w = i + \frac{p}{\rho}$

$$\frac{\partial w}{\partial t} + v \frac{\partial w}{\partial x} - \frac{1}{\rho} \left(\frac{\partial p}{\partial t} + v \frac{\partial p}{\partial x} \right) = \frac{1}{\rho} \left[\Lambda'_e - v \Lambda'_v - \left(w - \frac{v^2}{2} \right) \Lambda_\rho \right] \quad (13)$$

Introducing Gruneisen parameter ϕ defined as:

$$\phi = \frac{1}{\rho c_v} \left(\frac{\partial p}{\partial T} \right)_\rho = \frac{1}{\rho} \left(\frac{\partial p}{\partial i} \right)_\rho \quad (14)$$

Implementation of the second equation of the final system (4)



Then:

$$d\rho = \frac{1 + \phi}{c^2} dp - \frac{\rho\phi}{c^2} dw \quad (15)$$

$$\frac{\partial p}{\partial t} + v \frac{\partial p}{\partial x} = \frac{\rho\phi}{1 + \phi} \left(\frac{\partial w}{\partial t} + v \frac{\partial w}{\partial x} \right) - \frac{\rho c^2}{1 + \phi} \frac{\partial v}{\partial x} + \Lambda_\rho \frac{c^2}{1 + \phi} \quad (16)$$

Substituting (16) in (13) the final equation is obtained:

$$\frac{\partial p}{\partial t} + v \frac{\partial p}{\partial x} + \rho c^2 \frac{\partial v}{\partial x} = \phi \left[\Lambda'_e - v \Lambda'_v - \left(w - \frac{v^2}{2} - \frac{c^2}{\phi} \right) \Lambda_\rho \right] \quad (17)$$

Implementation of the third equation of the final system



Expliciting i in term of fluid temperature:

$$di = \left(\frac{p}{\rho} - \phi c_v T \right) \frac{d\rho}{\rho} + c_v dT \quad (18)$$

Substituting (18) in (9) and collecting terms:

$$\left(\frac{p}{\rho} - \phi c_v T \right) \left(\frac{\partial \rho}{\partial t} + v \frac{\partial \rho}{\partial x} \right) + \rho c_v \frac{\partial T}{\partial t} + \rho v c_v \frac{\partial T}{\partial x} + p \frac{\partial v}{\partial x} = \Lambda'_e - v \Lambda'_v - \left(i - \frac{v^2}{2} \right) \Lambda_\rho \quad (19)$$

$\searrow$
 $-\rho \frac{\partial v}{\partial x} + \Lambda_\rho$

From the mass conservation equation (1), the final equation is obtained:

$$\frac{\partial T}{\partial t} + v \frac{\partial T}{\partial x} + \phi T \frac{\partial v}{\partial x} = \frac{1}{\rho c_v} \left[\Lambda'_e - v \Lambda'_v - \left(w - \frac{v^2}{2} - \phi c_v T \right) \Lambda_\rho \right] \quad (20)$$